

$$\text{NL}_{\mathbf{k}_s \rightarrow 0} \left(\text{Diagram 1} \right) = \text{Diagram 2} \times \text{Diagram 3} \times \text{Diagram 4}$$

The diagram illustrates a factorization of a nonlinear interaction term $\text{NL}_{\mathbf{k}_s \rightarrow 0}$ into three components.

Diagram 1 (Left): A four-point interaction vertex (represented by a circle with a diagonal slash) is connected to four external square vertices. The external legs are labeled $\delta h_{\mathbf{k}_1}$, $\delta h_{\mathbf{k}_2}$, $\delta h_{\mathbf{k}_3}$, and $\delta h_{\mathbf{k}_4}$. Two internal wavy lines (representing gauge fields A^μ) connect the top and bottom vertices to the central vertex. The wavy lines are labeled $A^\mu_{\mathbf{q}}$ and $A^\mu_{\mathbf{q}+\mathbf{k}_1}$.

Diagram 2 (Middle): A four-point interaction vertex (represented by a circle with a cross) is connected to four external square vertices. The wavy lines are vertical and labeled $A^\mu_{\mathbf{q}}$ and $A^\mu_{\mathbf{q}+\mathbf{k}_1}$.

Diagram 3 (Right): A loop diagram consisting of two wavy lines forming a closed loop, with two external vertices (represented by black dots) connected to the loop.

Diagram 4 (Far Right): A four-point interaction vertex (represented by a circle with a cross) is connected to four external square vertices.